Uppsala University

Department of Information Technology

Artificial Intelligence

Instructions: Read through the complete exam and note below any unclear directives before you start solving the questions. **If you find any unclear directives, then please note the question number and explain what you think is unclear.**

This is a **closed book** exam. You may use a standard English dictionary and a calculator. You are NOT allowed to use any other aids (e.g. course material, notes, mobile phones). The exam is **individual**: you are NOT allowed to get help from other people to answer the questions. Cheating will be reported in accordance with university regulations.

Grading scale. 0-49 pts. (U), 50-69 pts. (3), 70-84 pts. (4), 85-100 pts. (5)

There are 15 multiple-choice questions, each worth 2 points (30 points total). There are 7 problem-style questions, each worth 10 points (70 points total), some of which are divided into 2 parts.

Fill in your answer here

1 In the context of artificial intelligence, what is an agent?
Select one alternative:

- Anything that can perceive its environment and act upon it
- A computer program that only stores information
- A human who interacts with machines
- A mechanical robot with physical sensors
- A central processing unit in a computer

Maximum marks: 2

2 Which of these is NOT one of the four AI approaches to rational agents?
Select one alternative:

- Acting emotionally
- Thinking rationally
- Thinking humanly
- Acting humanly
- Acting rationally

Maximum marks: 2

3 Which of the following statements is true about Shortest Path Problems?
Select one alternative:

- Can always be solved with BFS
- Can only be solved using A*
- Can only be solved if the graph is acyclic
- Can be used to find a route that minimises some cost
- Are only applied to graphs that have costs associated with the nodes

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Maximum marks: 2

4 A certain stock price has been observed to follow a pattern.

- If the stock price goes up one day, there's a 20% chance of it rising tomorrow, a 30% chance of it falling, and a 50% chance of it remaining the same.
- If the stock price falls one day, there's a 35% chance of it rising tomorrow, a 50% chance of it falling, and a 15% chance of it remaining the same.
- Finally, if the price is stable on one day, then it has a 50-50 chance of rising or falling the next day.

Which matrix below is the transition matrix for this Markov chain, if we list states in the order: (rising, falling, constant).

Select one alternative:

- None of the other options
- $\begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.35 & 0.5 & 0.15 \\ 0.5 & 0.5 & 0 \end{bmatrix}$
- $\begin{bmatrix} 20 & 30 & 50 \\ 35 & 50 & 15 \\ 50 & 50 & 0 \end{bmatrix}$
- $\begin{bmatrix} 20 & 35 & 50 \\ 30 & 50 & 50 \\ 50 & 15 & 0 \end{bmatrix}$
- $\begin{bmatrix} 0.2 & 0.35 & 0.5 \\ 0.3 & 0.5 & 0.5 \\ 0.5 & 0.15 & 0 \end{bmatrix}$

Maximum marks: 2

5 In **forward state-space search** planning, what does the planner primarily expand during search?
Select one alternative:

- Goal regressions derived from the final state
- Partial-order plans represented as DAGs
- States reachable from the initial state through applicable actions
- Sets of possible world states under uncertainty
- Sequences of actions that achieve partial goals

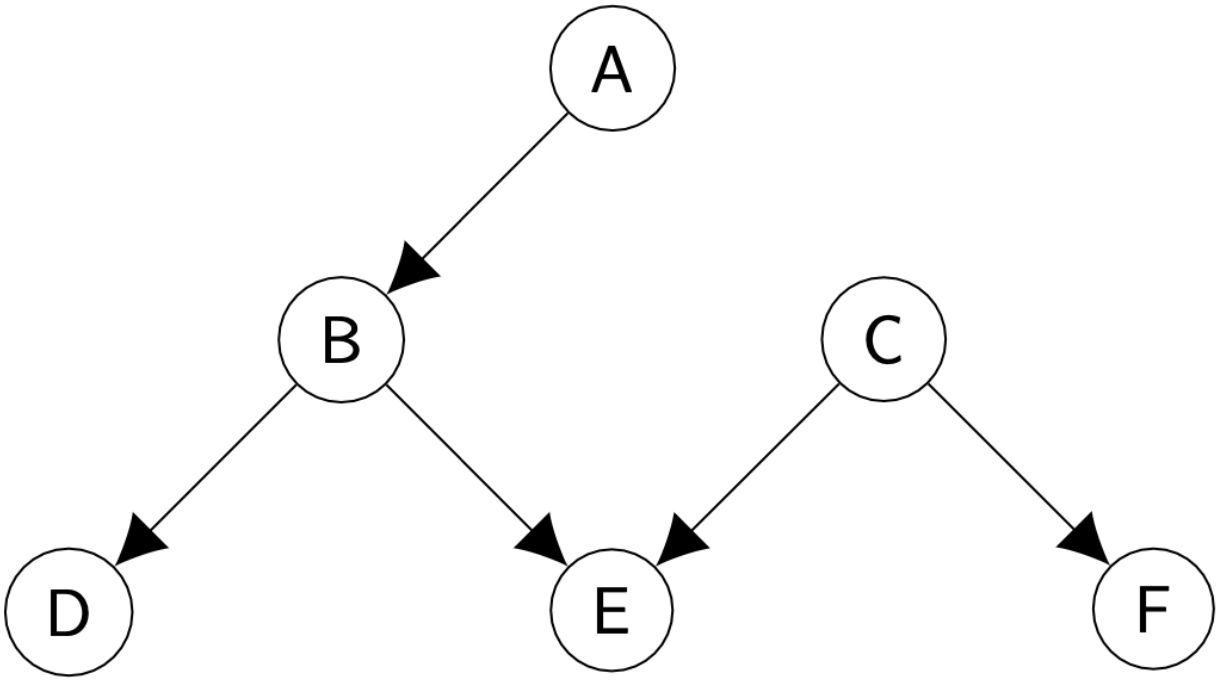
Maximum marks: 2

6 Which of the following is **not** a characteristic of alpha-beta pruning?
Select one alternative:

- It is a greedy algorithm
- It is used in game theory
- It is a recursive algorithm
- It can only be used in two-player games with perfect information
- It is used in decision making

Maximum marks: 2

7 Consider the following bayesian network:



Which probability distribution is associated with node B?
Select one alternative:

- P(B|A)
- P(B|A,D,E)
- P(A|B)
- P(B)
- P(B|D,E)

Maximum marks: 2

8 What is Viterbi's algorithm used for in Hidden Markov Models?
Select one alternative:

- To compute the likelihood of an observed sequence given the model parameters.
- To generate random sequences of observations based on the HMM parameters.
- To estimate the transition and emission probabilities of the HMM using observed data.
- To cluster observed sequences into groups based on similarity without considering hidden states.
- To find the most probable sequence of hidden states that could have generated the observed sequence.

Maximum marks: 2

9 Which of the following is **not** a component of a Hidden Markov Model?
Select one alternative:

- All of the other alternatives are part of a HMM
- Emission probabilities
- Decision Tree
- Initial state probabilities
- State transition probabilities

Maximum marks: 2

10 One of the main goals of AI in video games (as opposed to academic AI) is:
Select one alternative:

- To control the colour scheme of the game based on the player's facial expressions
- To create opponents with believable behaviour
- To design the game's graphics
- To create unbeatable opponents
- To study optimal game strategies

Maximum marks: 2

11 Which algorithm is inspired by the social behaviour of bird flocking or fish schooling?
Select one alternative:

- Simulated Annealing
- Hill Climbing
- Genetic Algorithm
- Automated Bee Colony Domain Search
- Particle Swarm Optimisation

Maximum marks: 2

12 What happens when the heuristic function in the A* algorithm underestimates the cost?
Select one alternative:

- The algorithm becomes faster
- The algorithm finds a shortest path using tree search
- The algorithm becomes uninformed
- The algorithm becomes slower
- The algorithm becomes non-optimal

Maximum marks: 2

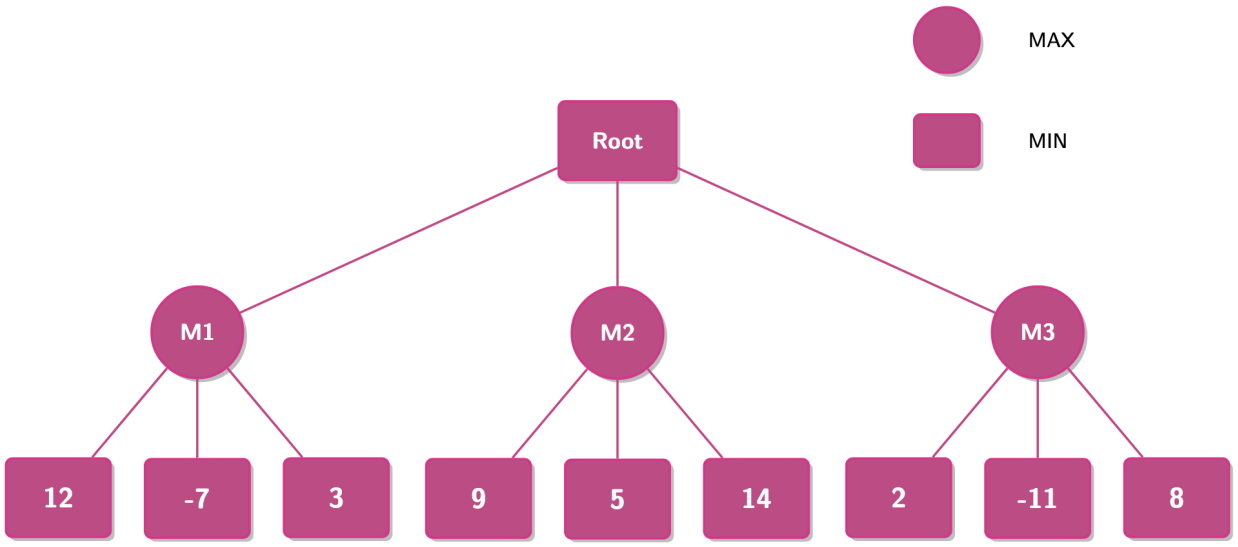
13 What is the primary concept behind Ant Colony Optimisation?
Select one alternative:

- Simulating the behaviour of ants to build complex structures
- Using ant behaviour to predict weather patterns
- Simulating the behaviour of ants to find the shortest path
- Using ant behaviour to solve complex mathematical equations
- Modelling the behaviour of ants to create realistic video game characters

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Maximum marks: 2

14 What would be the evaluation value for the **MIN** player if a *minimax* algorithm is applied on the following game tree?



Select one alternative:

- 2
- 5
- -11
- 3
- 8

Maximum marks: 2

15 When extending **minimax** to a **three-player** game, what is the most direct conceptual generalization?

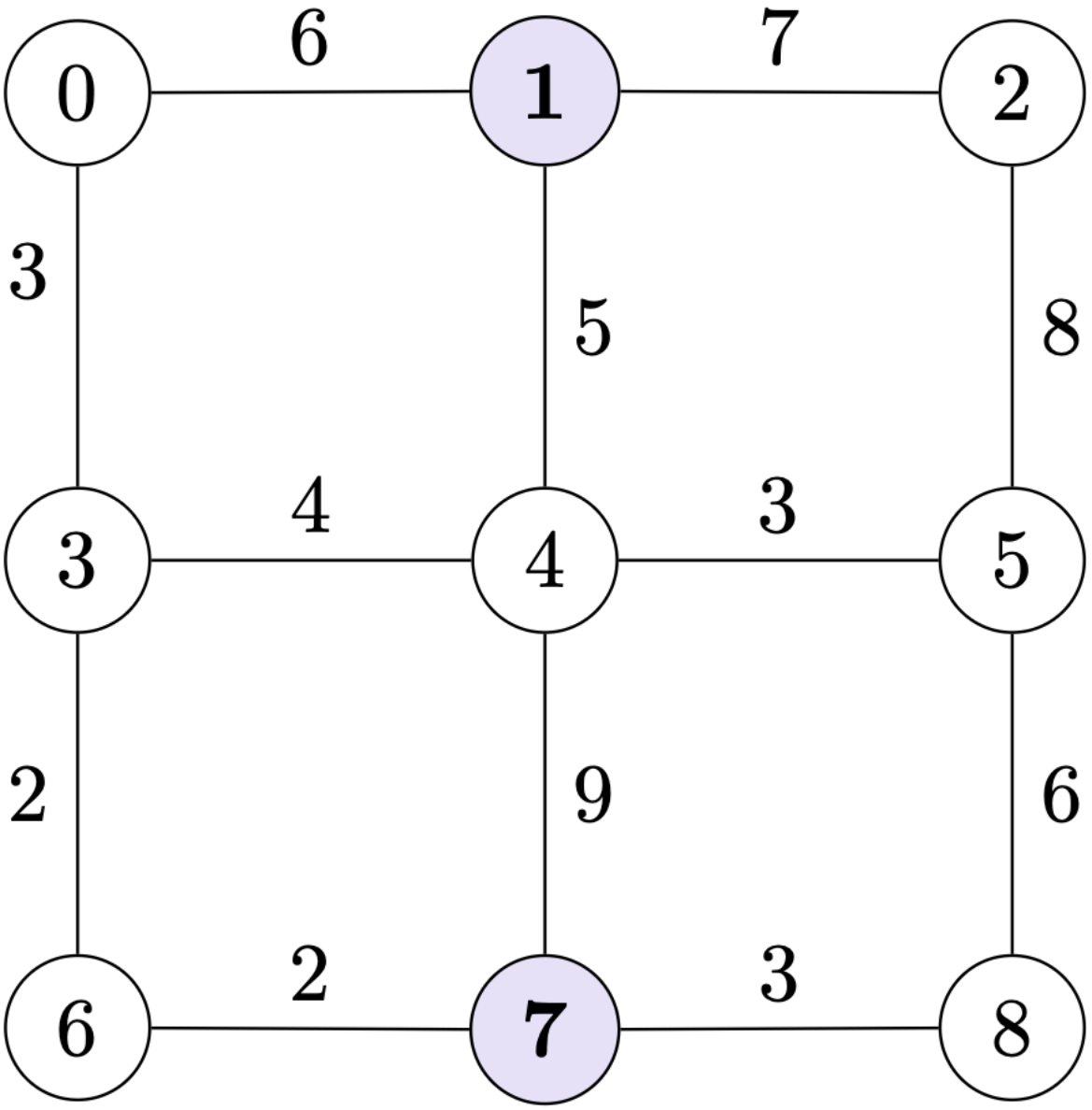
Select one alternative:

- Each player’s move is chosen randomly to maintain fairness
- Only two players are considered at any given time, with the third ignored
- Each player maximizes the same utility function
- The algorithm assigns alternating MAX and MIN layers as in the two-player case
- Players alternate turns, and each maximizes their own utility

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Maximum marks: 2

16 Use A* to find a shortest path from the start node 1 to the goal node 7 in the graph below:



Provide an admissible heuristic (2 points)

Fill in your answer here

Show all your workings (all the steps in the algorithm to find the goal) (8 points)

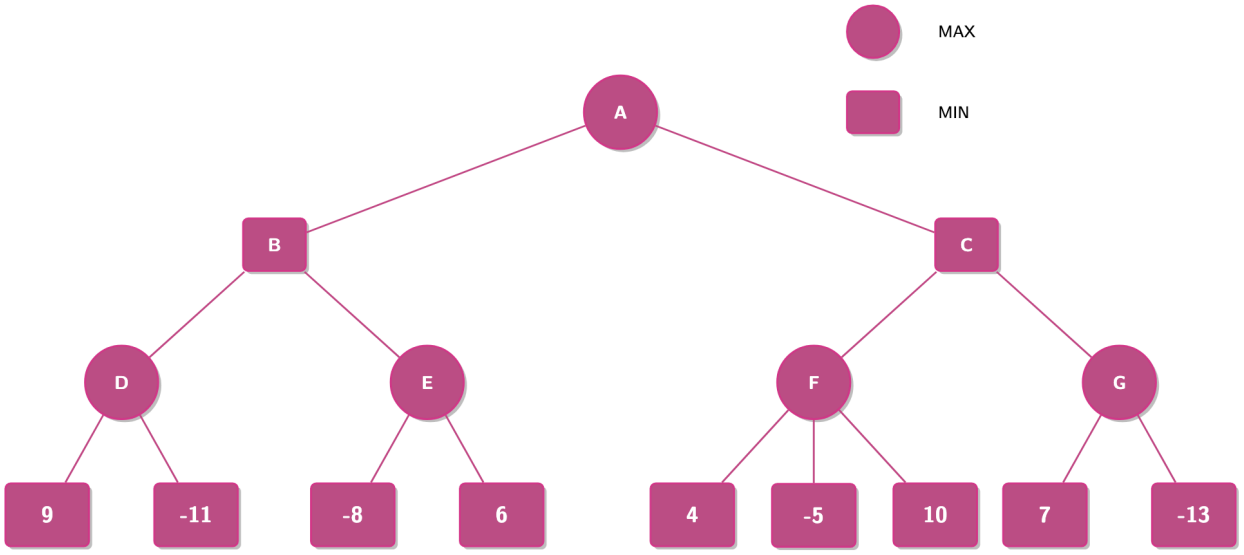
Fill in your answer here

Maximum marks: 10

17 Perform alpha-beta pruning on this game tree assuming that branches are explored from left to right. You should show all workings, where this means all alpha-beta values (see example below). Provide a way of identifying the sequence of updates to alpha-beta values associated with nodes. Use the node indices in the diagram as a way of identifying which particular alpha-beta values are associated with which node . Show where pruning occurs. Finally, provide the result (value at end state) of the game assuming optimal play.

For example, use the following notation:

Step X: Node A [alpha, beta]
Step Y: Node B return value
Step Z: Node C because 1<2 then ...



Fill in your answer here

Maximum marks: 10

18 Consider the actions in the table below:

Index	Action	Duration	Uses	Consumes	After
0	Start	0			NA
1	Action 1	33	Pencil	-1 draft art	0
2	Action 2	54	Paint	-1 draft art	0
3	Action 3	22	Brush		1
4	Action 4	46			2,3
5	Action 5	11	Paint, Brush	1 draft art	2,3
6	Action 6	37	Pencil		4,5
7	Action 7	28	Paint, Brush	1 draft art	5
8	Finish	0			4,6,7

Provide a complete unconstrained schedule (5 points)

Fill in your answer here

Provide a complete resource-constrained schedule (5 points)

Fill in your answer here

Maximum marks: 10

19 What is PDDL? Name and explain all components of a PDDL problem. Be as precise as possible and give a small example. You do not have to worry about the exact syntax of PDDL expressions, but you should have all the components of PDDL expressions. In your small example explain what each syntactic element does. (10 points)

Fill in your answer here

Maximum marks: 10

20 Write an essay about the definition artificial intelligence (5 points) and the history of artificial intelligence (5 points). Name at least five important periods or events and provide approximate dates for them (10 points total).

Fill in your answer here

Maximum marks: 10

21 You do not need to use a calculator for this question. All your answers should be algebraic or symbolic expressions. You do not need to use the equation editor, your equations can be in ASCII or something approximating LaTeX, but you are free to use the equation editor if you want.

You have a pet rabbit called Minibean. She is a rather stoic creature who does not show her emotions very well. Also, She does not lead a rich inner life. Minibean has only two mental states: Happy and Sad Nonetheless, as we will see later these mental states affect her behaviour.

Question A (1 point): Using an MRI machine and a deep neural network you are able to determine if Minibean is happy or sad. This is a costly procedure, but after some experimentation you have estimates of the following quantities. First the probability that Minibean is happy or sad $P(H), P(S)$, and second the transition probabilities of Minibean's mental state on the next day given her mental state on the previous day: $P(H|H), P(H|S), P(S|H)$ and $P(S|S)$. Write the expression for the probability Minibean is sad after three days given the initial probabilities $P(H_0), P(S_0)$.

Question B (2 points): Using the MRI machine is costly, and you do not always have access to it, but Minibean's mental state affects what she eats. If she is sad then there is a higher probability that she eats carrots (high sugar and instance gratification), if she is happy then there is a higher probability that she will munch slowly on Alfalfa. Without the MRI machine you are going to model Minibean's behaviour using a Hidden Markov model (HMM). With hidden states; Happy H and Sad S. You observe the rabbit eating Carrots C or Alfalfa A. To model this as a HMM you need a number of other emission probabilities. Specify what probabilities you need and what they mean.

Question C (1 point): Suppose that you knew Minibean's mental state for 3 days is H, H, H. Write an expression for the probability that she eats Alfalfa on the third day.

Question D (2 points): In general you cannot observe the mental state of Minibean, so instead you are going to take note of her eating patterns and try to infer her mental state. Suppose that you observe the sequence C, A, there are two approaches to calculating the probability of her mental states. One way is to consider all possible sequences of hidden states. Write an expression for the probability that Minibean is happy given the observed sequence. Explain and justify why this expression is correct.

Question E (4 points): Continuing from part D. Enumerating all the possible sequences of hidden states is in general too costly computationally. There is another algorithm that is much more efficient. For the observed sequence in part D, show how you could calculate the probability that Minibean is happy. You need to explain and justify why you answer is correct.

Fill in your answer here

Maximum marks: 10

22 You have a job as an ethical and legal adviser for a major company that builds smart TVs. The company is planning to include advanced AI in the design. At the moment, there are lots of ideas floating around on how to use AI to improve the customer experience.

The marketing department believes the TV should recommend new content to users based on their past behaviour. The engineering department thinks that it is possible to use speech recognition to listen to conversations and analyze what people are interested in. To make this system more useful, it will work out who is speaking by their voice patterns and build an AI model of each person in the household. This AI model will be used to drive recommendations. The marketing company also thinks it will be able to sell this information to a video streaming service so that it can make content that people want to watch.

Your boss thinks that there will be no legal problems with the service as users will have to agree to terms of service when they use the TV.

Discuss how this system fits into the European Framework for Trustworthy AI and the seven requirements for a trustworthy system. The seven requirements are:

1. Human Agency and Oversight
2. Technical Robustness and Safety
3. Privacy and Data Governance
4. Transparency
5. Diversity, Non-discrimination and Fairness
6. Environmental and Societal Well-Being
7. Accountability

Discuss each of the seven requirements and possible problems and violations that could occur with the new AI-driven TV.

Fill in your answer here

Maximum marks: 10